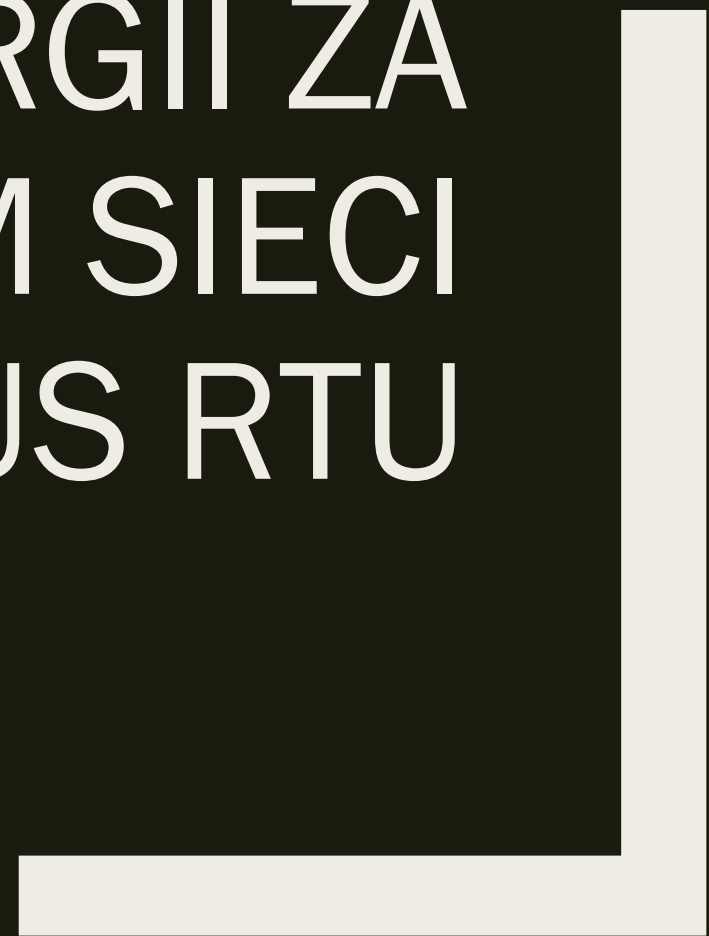




INTERNET RZECZY

Patryk Choiński
Rafał Celiński

SYSTEM ODCZYTU LICZNIKÓW ENERGII ZA POŚREDNICTWEM SIECI MODBUS RTU



INTERPRETACJA TEMATU

Miernik Energii Delta D520i



Modbus RTU

PLC Delta AX8



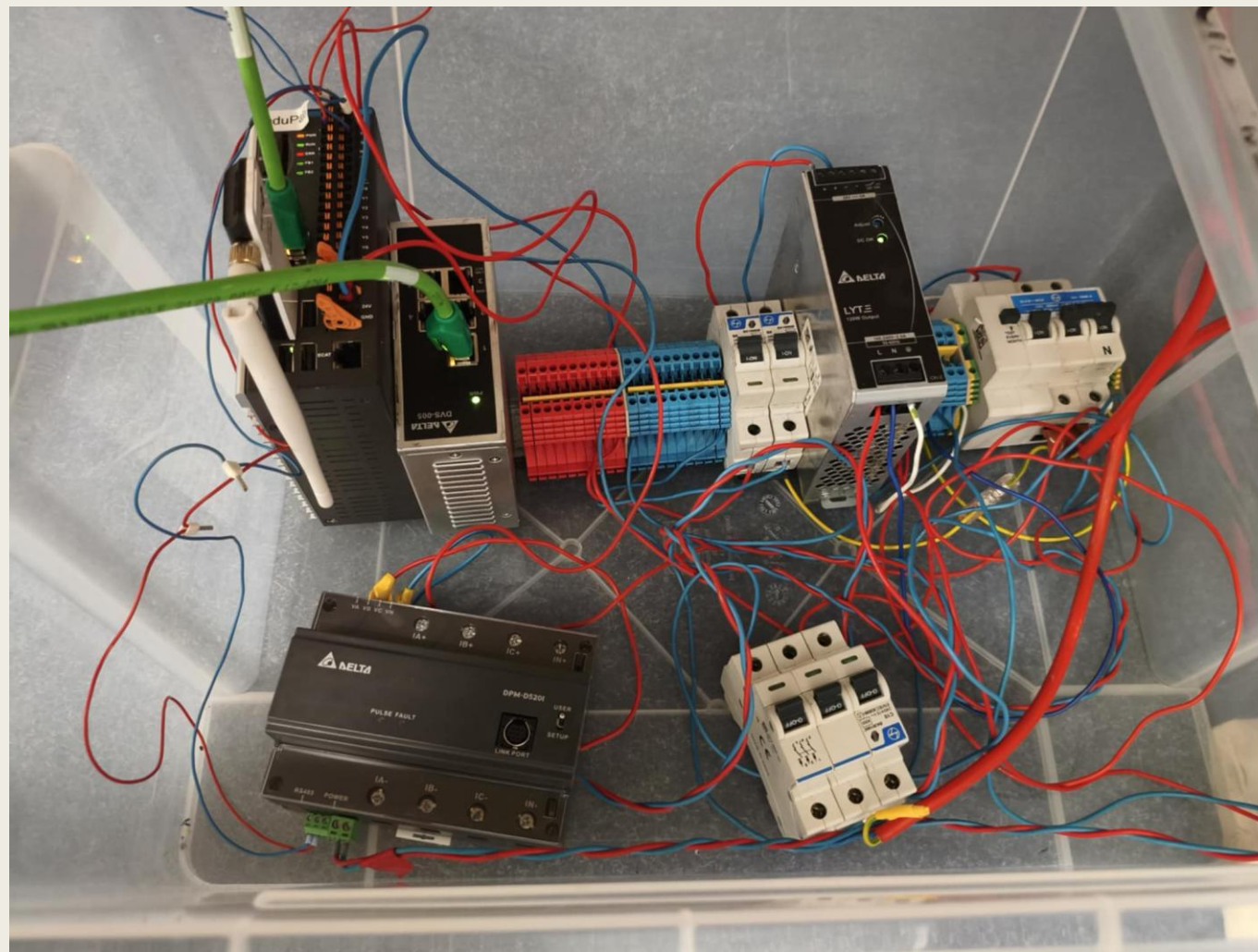
NodeRed Server



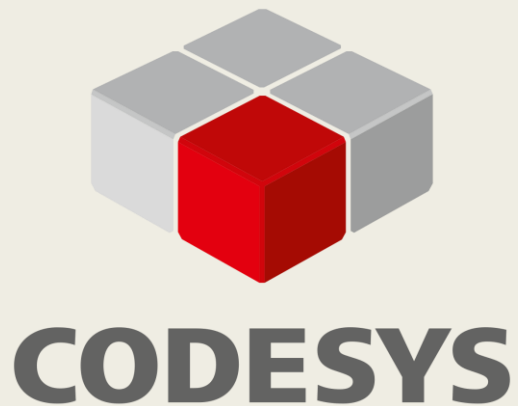
OPC UA

OpenVPN

ZESTAWIENIE SPRZĘTU



TECHNOLOGIE



Publiczne Repozytorium

- <https://github.com/PatrykChoinski/IoT-NodeRed-EnergyMeter>

The screenshot shows the GitHub repository page for **IoT-NodeRed-EnergyMeter** by user **PatrykChoinski**. The repository is public and has 2 issues, 0 pull requests, 0 projects, 0 wiki pages, 0 security advisories, 0 insights, and 0 settings. The repository has 0 stars, 0 forks, and 0 watchers. The repository is currently on the **master** branch, with 2 branches and 0 tags. The repository contains 131 commits.

The repository files and their commit history are listed below:

File	Commit Message	Commit Date
.github/workflows	yml	last week
Documentation	Docs, Readme PLC	3 weeks ago
PLC	Merge branch 'develop/Nodered'	yesterday
.flows.json.backup	Zmiana rozmieszczen elementow i poprawy nazwenictwa	yesterday
.gitignore	Update Gitignore	yesterday
README.md	Update README.md	last week
flows.json	Zmiana rozmieszczen elementow i poprawy nazwenictwa	yesterday
flows_cred.json	Powrot bez folderu ./nodered	last week
package-lock.json	ij	3 weeks ago
package.json	...	last week
server.js	Update server.js	last week
settings.js	Update server.js	last week

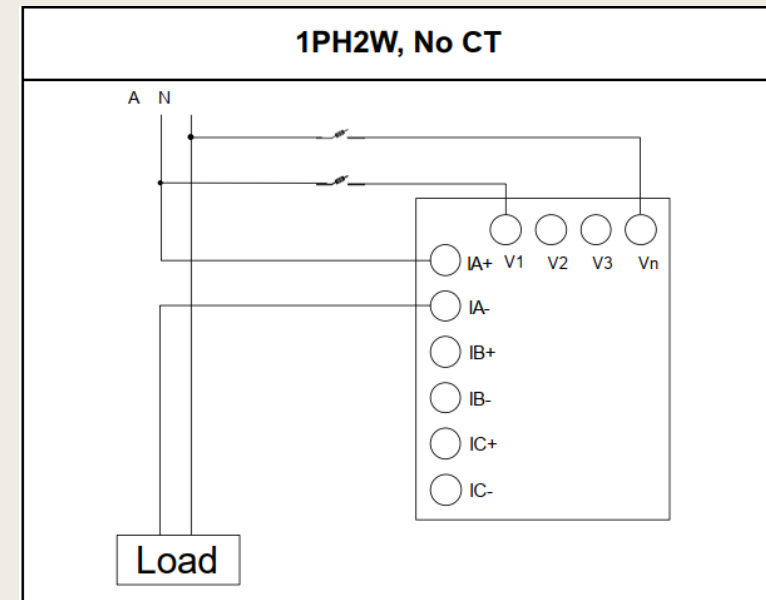
The repository also includes an **About** section with no description, website, or topics provided. It also has **Releases** and **Packages** sections, both of which are empty. The **Contributors** section lists 2 contributors: **rafcel2222-prog** and **PatrykChoinski** (Patryk Choiński).

Podłączenie Miernika

- Mierzymy prąd i napięcie z odbiornika, którym jest zasilacz 230VAC->24VDC. Miernik nie potrzebuje przekładnika CT.



Zasilanie



MODBUS RTU

Slave



Modbus RTU

Serial Port Configuration

COM port	3
Baud rate	9600
Parity	NONE
Data bits	8
Stop bits	1

Master



MODBUS RTU

Dokumentacja

1. Meter Parameters: 0100 – 01FF							
100	40257	Phase A voltage	0.000 – 99999.999	float	V	4	R
101	40258						
102	40259	Phase B voltage	0.000 – 99999.999	float	V	4	R
103	40260						
104	40261	Phase C voltage	0.000 – 99999.999	float	V	4	R
105	40262						
106	40263	Average phase voltage	0.000 – 99999.999	float	V	4	R
107	40264						
108	40265	A–B line voltage	0.000 – 99999.999	float	V	4	R
109	40266						
10A	40267	B–C line voltage	0.000 – 99999.999	float	V	4	R
10B	40268						
10C	40269	C–A line voltage	0.000 – 99999.999	float	V	4	R
10D	40270						
10E	40271	Average line voltage	0.000 – 99999.999	float	V	4	R
10F	40272						
110	40273	Phase A voltage unbalance	0.00 – 99.99	float	%	4	R
111	40274						
112	40275	Phase B voltage unbalance	0.00 – 99.99	float	%	4	R
113	40276						

Cykliczne czytanie paramtertów funkcja Read Holding Registers

	Name	Access Type	Trigger	READ Offset	Length	Error Handling
0	VoltageA	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0100	2	Keep last value
1	CurrentA	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0120	2	Keep last value
2	PowerFactA	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0134	2	Keep last value
3	Frequency	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0142	2	Keep last value
4	ActivePowerA	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0146	2	Keep last value
5	ReactivePowerA	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#014C	2	Keep last value
6	ApparentPowerA	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0156	2	Keep last value
7	THDCA	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0174	2	Keep last value
8	THDVA	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#017C	2	Keep last value
9	MaxVoltageA	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0212	6	Keep last value
10	MaxCurrentA	Read Holding Registers (Function Code 03)	Cyclic, t#100ms	16#0224	6	Keep last value

Variable	Mapping	Channel	Address	Type	Current Value
 Application.VoltageRaw		VoltageA	%IW0	ARRAY ...	[53084,17243]
 Application.CurrentRaw		CurrentA	%IW2	ARRAY ...	[52953,15863]
 Application.PowerFactorRaw		PowerF...	%IW4	ARRAY ...	[65535,65535]
 Application.FrequencyRaw		Frequency	%IW6	ARRAY ...	[4404,16968]
 Application.ActivePowerRaw		ActiveP...	%IW8	ARRAY ...	[55050,48163]
 Application.ReactivePowerRaw		Reactiv...	%IW10	ARRAY ...	[39846,15556]
 Application.ApparentPowerRaw		Appare...	%IW12	ARRAY ...	[12059,15581]
 Application.THDCARaw		THDCA	%IW14	ARRAY ...	[50987,17045]
 Application.THDVARaw		THDVA	%IW16	ARRAY ...	[51905,16033]
 Application.MaxVoltageRaw		MaxVolt...	%IW18	ARRAY ...	[46268,17258,6411,2,1842,17]
 Application.MaxCurrentRaw		MaxCur...	%IW24	ARRAY ...	[36176,15895,6411,3,2817,4]

MODBUS 2x Word -> Float. Jak to zrobić?

Variable	Mapping	Channel	Address	Type	Current Value
Application.VoltageRaw		VoltageA	%IW0	ARRAY ...	[53084,17243]
Application.CurrentRaw		CurrentA	%IW2	ARRAY ...	[52953,15863]
Application.PowerFactorRaw		PowerF...	%IW4	ARRAY ...	[65535,65535]
Application.FrequencyRaw		Frequency	%IW6	ARRAY ...	[4404,16968]
Application.ActivePowerRaw		ActiveP...	%IW8	ARRAY ...	[55050,48163]
Application.ReactivePowerRaw		Reactiv...	%IW10	ARRAY ...	[39846,15556]
Application.ApparentPowerRaw		Appare...	%IW12	ARRAY ...	[12059,15581]
Application.THDCARaw		THDCA	%IW14	ARRAY ...	[50987,17045]
Application.THDCARaw		THDVA	%IW16	ARRAY ...	[51905,16033]
Application.MaxVoltageRaw		MaxVolt...	%IW18	ARRAY ...	[46268,17258,6411,2,1842,17]
Application.MaxCurrentRaw		MaxCur...	%IW24	ARRAY ...	[36176,15895,6411,3,2817,4]

Oczywiście użycie Unii

https://en.wikipedia.org/wiki/Union_type

```
TYPE WORDS_TO_REAL :  
  UNION  
    Words : ARRAY [0..1] OF WORD;  // Dwa słowa po 16 bitów  
    Value : REAL;                   // Ta sama pamięć widziana jako REAL (32 bity)  
  END_UNION  
END_TYPE
```

```
// Napięcie  
● Converter.Words[0] 36176 := VoltageRaw[0] 27329;  
● Converter.Words[1] 15895 := VoltageRaw[1] 17245;  
● Voltage 221 ▶ := Converter.Value 0.148;
```

Prepare Value

Expression Voltage

Type REAL

Current value 221.382

What do you want to do?

☒ Prepare a new value for the next write or force operation

221.382

☐ Remove preparation with a value.

☐ Release the force, without modifying the value.

☐ Release the force and restore the variable to the value it had before forcing it.

OK Cancel

OPC UA

Add Symbol Configuration

Create a remote access symbol configuration.

Name

Symbol Configuration

☒ Include comments in XML

☒ Support OPC UA features

☐ Add library placeholder in Device Application
(recommended, but may trigger download)

Client Side Data Layout

☐ Compatibility Layout

☒ Optimized Layout

Add

Cancel

Symbols	Access Rights	Maximal	Attribute	Type	Members
Constants					
GVL					
ActivePower				REAL	
ApparentPower				REAL	
Current				REAL	
Frequency				REAL	
MaxCurrent				REAL	
MaxCurrentDay				WORD	
MaxCurrentHour				BYTE	
MaxCurrentMinute				BYTE	
MaxCurrentMonth				BYTE	
MaxCurrentSecond				WORD	
MaxCurrentYear				BYTE	
MaxVoltage				REAL	
MaxVoltageDay				WORD	
MaxVoltageHour				BYTE	
MaxVoltageMinute				BYTE	
MaxVoltageMonth				BYTE	
MaxVoltageSecond				WORD	
MaxVoltageYear				BYTE	
PowerFactor				REAL	
ReactivePower				REAL	
THDCA				REAL	
THDVA				REAL	
Voltage				REAL	
IoConfig_Globals					
ModbusRawData					
PLC_PRG					

OPC UA – sprawdzenie OPC WATCH

OPC Watch

TRAEGER.de

Server Browser

(1) opc.tcp://10.12.0.14:4840/

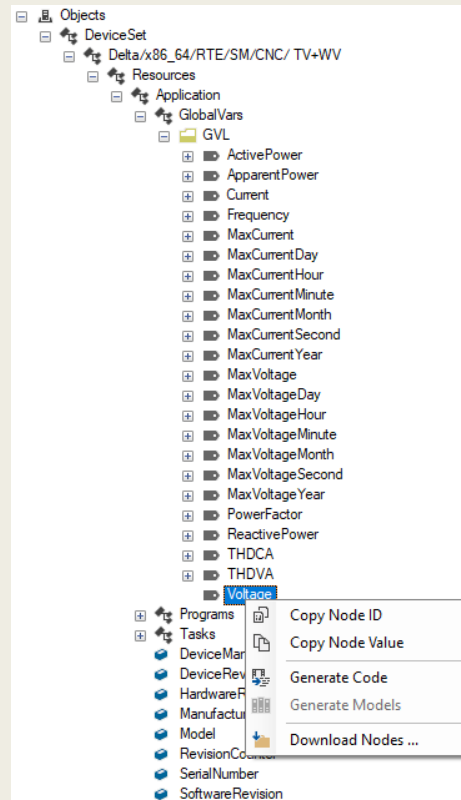
- Objects
 - DeviceSet
 - Delta/x86_64/RTE/SM/CNC/ TV+VW
 - Resources
 - Application
 - GlobalVars
 - GVL
 - ActivePower
 - ApparentPower
 - Current
 - Frequency
 - MaxCurrent
 - MaxCurrentDay
 - MaxCurrentHour
 - MaxCurrentMinute
 - MaxCurrentMonth
 - MaxCurrentSecond
 - MaxCurrentYear
 - MaxVoltage
 - MaxVoltageDay
 - MaxVoltageHour
 - MaxVoltageMinute
 - MaxVoltageMonth
 - MaxVoltageSecond
 - MaxVoltageYear
 - PowerFactor
 - ReactivePower
 - THDCA
 - THDVA
 - Voltage
 - Programs
 - Tasks
 - DeviceManual
 - DeviceRevision
 - HardwareRevision
 - Manufacturer
 - Model
 - RevisionCounter
 - SerialNumber
 - SoftwareRevision
- Server
- Types
- Views

Configuration Monitor Analyse

Read Write Auto Update Copy Node ID (1) opc.tcp://10.12.0.14:4840/Objects/DeviceSet/Delta/x86_64/RTE/SM/CNC/ TV+VW/Resources/Application/GlobalVars/GVL

Name	Value	Type	Base Type	Status	Server Timestamp	Source Timestamp	AccessLevel	Description	NodeId
ActivePower	-0.01	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
ApparentPower	0.027	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
Current	0.122	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
Frequency	50.018	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxCurrent	0.148	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxCurrentDay	0	WORD	UInt16	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxCurrentHour	0	BYTE	Byte	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxCurrentMinute	0	BYTE	Byte	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxCurrentMonth	0	BYTE	Byte	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxCurrentSecond	0	WORD	UInt16	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxCurrentYear	0	BYTE	Byte	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxVoltage	234.706	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxVoltageDay	0	WORD	UInt16	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxVoltageHour	0	BYTE	Byte	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxVoltageMinute	0	BYTE	Byte	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxVoltageMonth	0	BYTE	Byte	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxVoltageSecond	0	WORD	UInt16	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
MaxVoltageYear	0	BYTE	Byte	Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
PowerFactor	NaN	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
ReactivePower	0.025	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
THDCA	74.517	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
THDVA	0.789	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...
Voltage	218.37	Float		Good - The ope...	4.11.2025 10:09:35	4.11.2025 10:09:35	CurrentReadOrWrite		ns=4;s=IvarDelta...

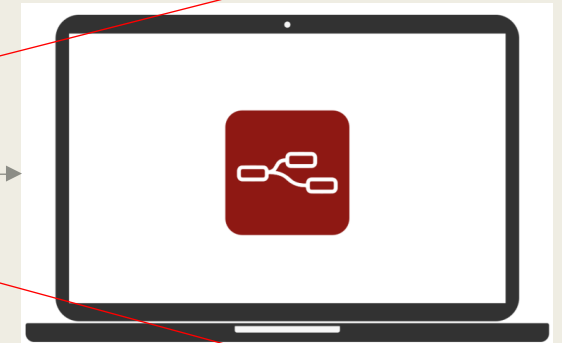
Dostęp do węzła zmiennej w OPC UA



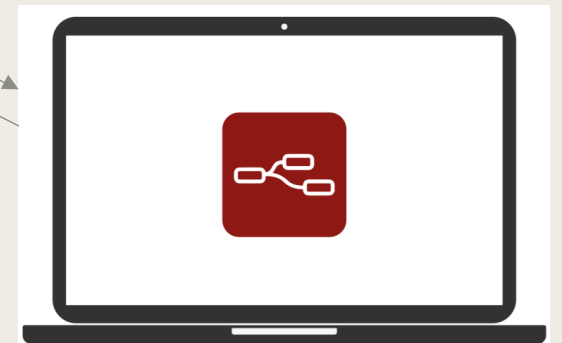
Konfiguracja OpenVPN

PLC Delta AX8

NodeRed Server



Ethernet



Konfiguracja OpenVPN

Ustawienia Servera

OpenVPN

☒ Włącz serwer OpenVPN

Dynamiczny adres IP: . . .

Maksymalna liczba połączeń:

Maksymalna liczba połączeń z kontem:

Port:

Protokół:

Szyfrowanie:

Uwierzytelnienie:

Wartość opcji mssfix:

☒ Włącz kompresję łącza VPN

☐ Pozwól klientom na dostęp do sieci LAN serwera [i](#)

☒ Sprawdź klucz uwierzytelniania TLS [i](#)

☐ Sprawdź CN serwera [i](#)

☐ Włącz tryb serwera IPv6

Prefiks:

[Eksportuj konfigurację](#)

Ustawienia Klienta PLC

BYTES IN 161 B/S ↓

↑ BYTES OUT 204 B/S

DURATION 15:59:46

PACKET RECEIVED 0 sec ago

YOU

_TestVPN_AX8

YOUR PRIVATE IP 10.12.0.14

SERVER

213.241.10.107

SERVER PUBLIC IP 213.241.10.107

PORT 31194

VPN PROTOCOL UDP

Ustawienia Klienta NodeRed

Securely Connected!
3d:16:04:58

 **213.241.10.107 PCHOINSKI**
213.241.10.107

[Disconnect](#)

[Hide Details](#)

You:

Private IP (IPv4) 10.12.0.6

Server:

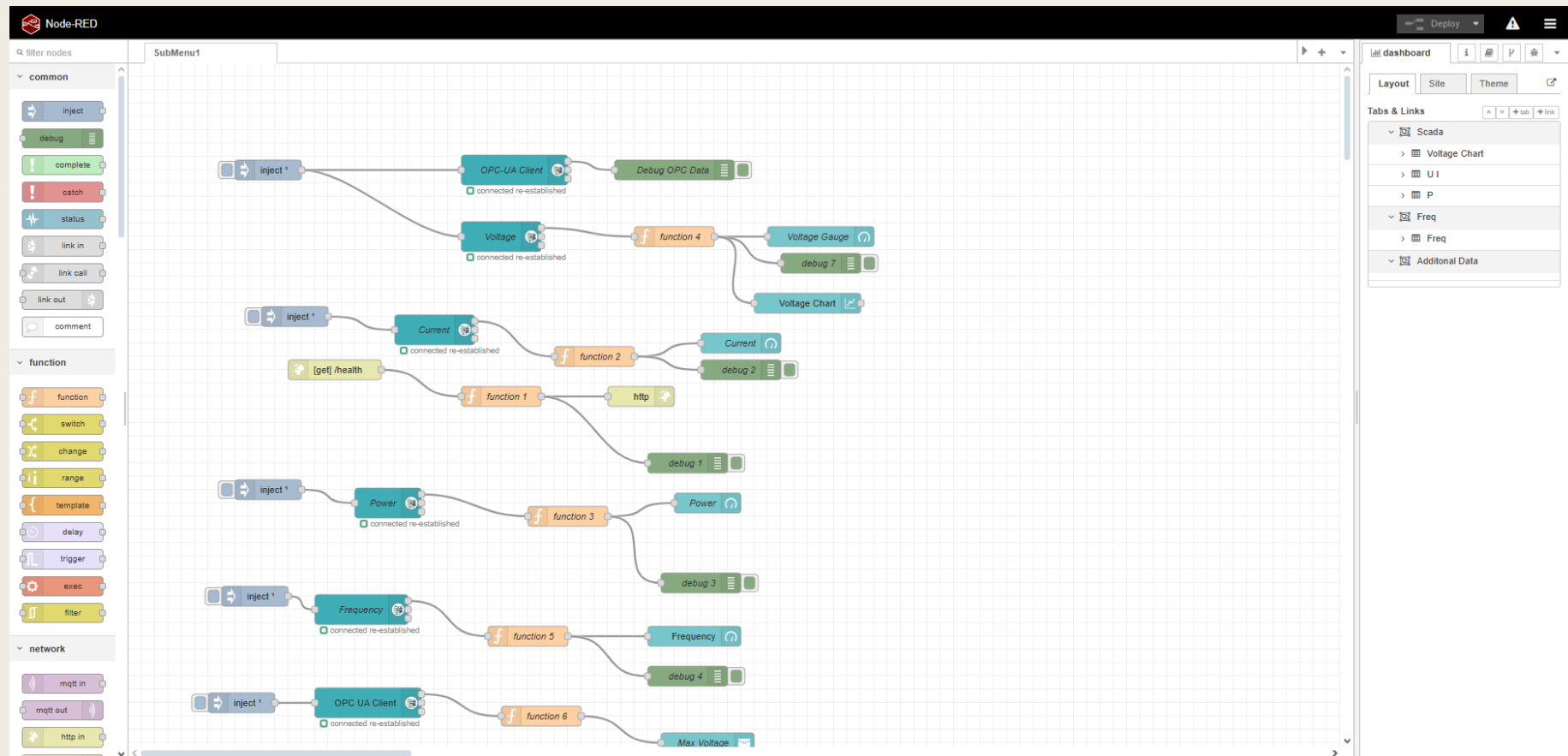
Server Public IP 213.241.10.107

Port / Protocol 31194 / UDP

Node-Red

<http://127.0.0.1:1880/#flow/>

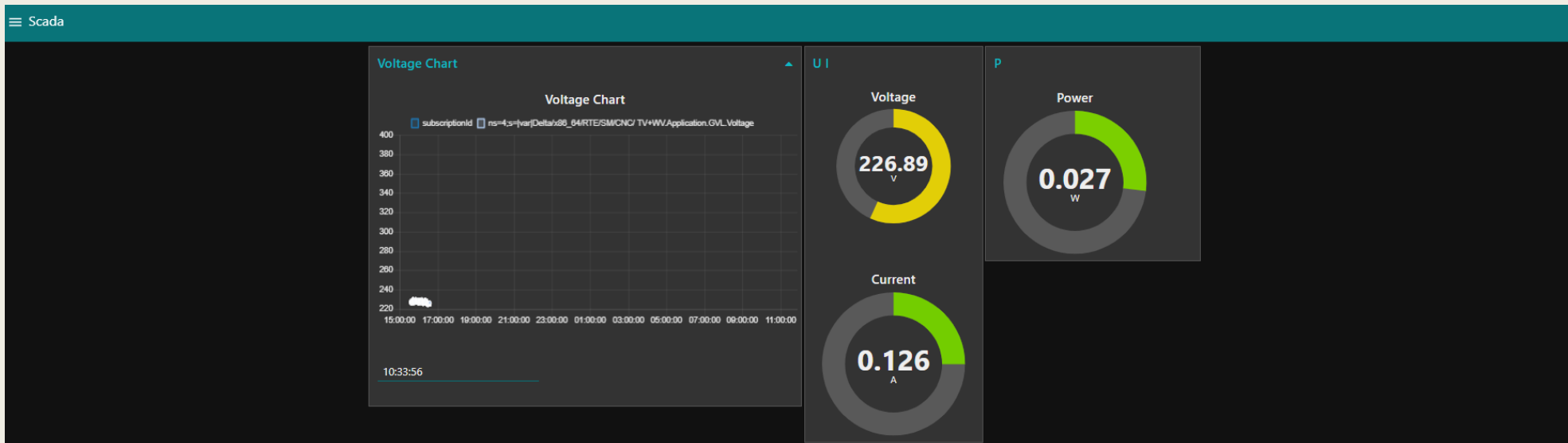
<http://127.0.0.1:1880/ui/>



Node-Red

<http://127.0.0.1:1880/#flow/>

<http://127.0.0.1:1880/ui/>

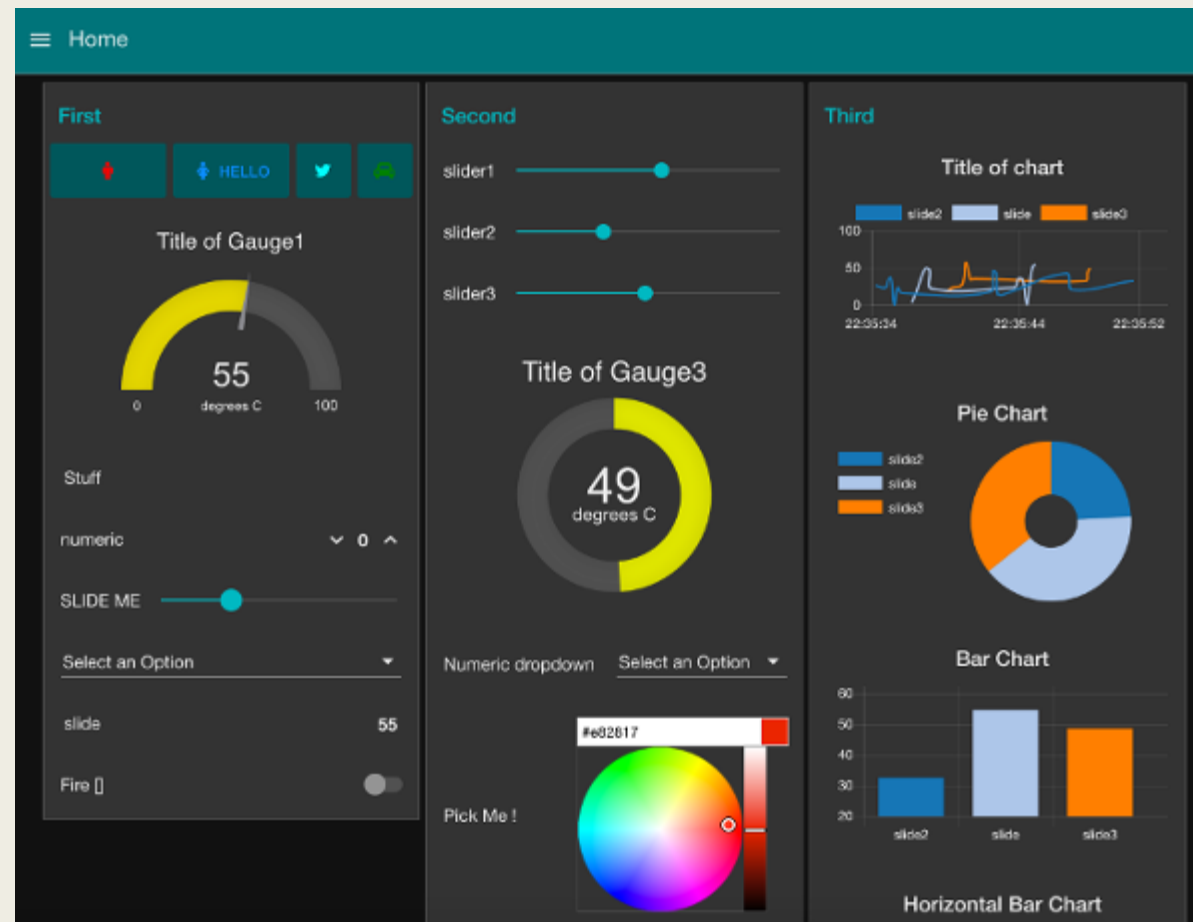


Node-Red – wykorzystane nodes

 node-red-contrib-alarm  ^1.4.0	remove from project
 node-red-contrib-github-plus  ^0.3.1	remove from project
 node-red-contrib-opcua  ^0.2.319	install
 node-red-contrib-web-worldmap  ^5.5.1	remove from project
 node-red-dashboard  ^3.6.0	

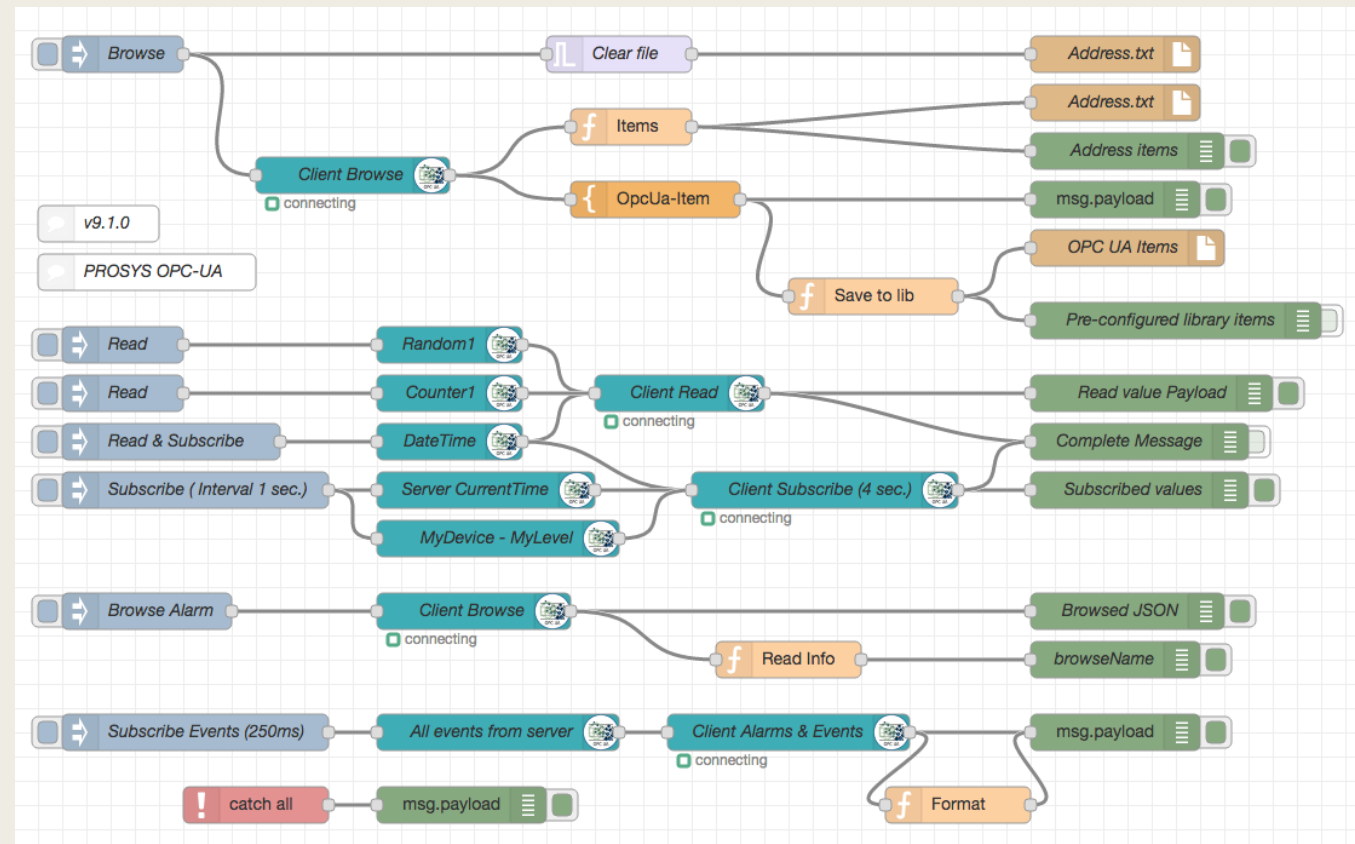
Node-Red Dashboard

<https://flows.nodered.org/node/node-red-dashboard>



Node-Red Opc-UA

<https://flows.nodered.org/node/node-red-contrib-opcua>



Node-Red Opc-UA Konfiguracja

Edit OpcUa-Client node > Edit OpcUa-Endpoint node

Delete Cancel Update

Properties

Endpoint

SecurityPolicy

SecurityMode

☒ Anonymous
☐ use credentials
☐ user certificate

Edit OpcUa-Client node

Delete Cancel Done

Properties

Endpoint

Action

Certificate

Local certificate file, fixed path

Local private key file, fixed path

Use transport settings ☐

Max ChunkCount

Max MessageSize

Receive BufferSize

Send BufferSize

Set statuscode and timestamp ☐

Keep session alive ☐

Name

Edit OpcUa-Client node

Delete Cancel Done

Properties

Endpoint

Action

Interval

Certificate

Local certificate file, fixed path

Local private key file, fixed path

Use transport settings ☐

Max ChunkCount

Max MessageSize

Receive BufferSize

Send BufferSize

Set statuscode and timestamp ☐

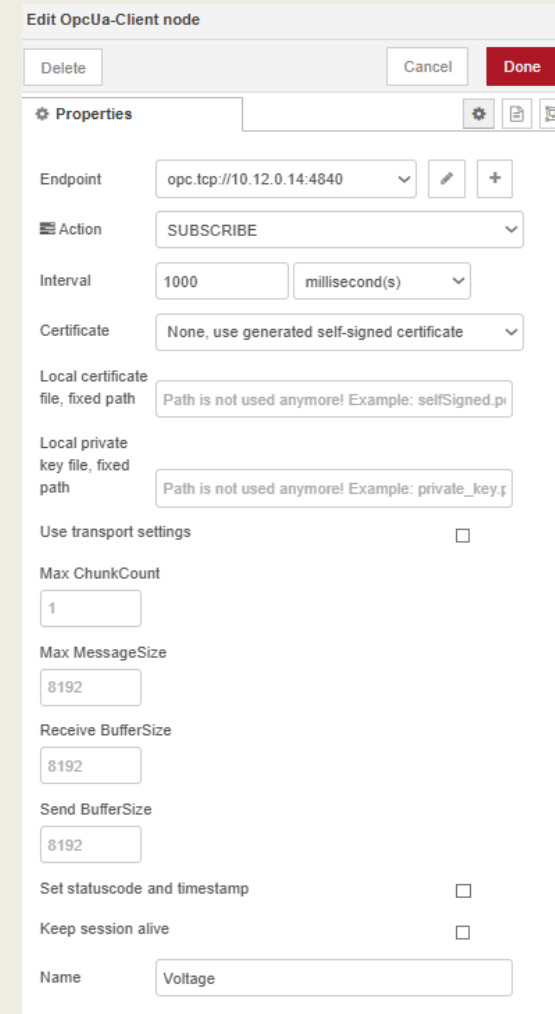
Keep session alive ☐

Name

Node-Red Opc-UA Subskrybcje

Zamiast wykorzystywać synchroniczne odpytywanie zmiennych możemy użyć subskrypcji(asynchronicznych), aby odciążyć ilość i częstotliwość wymiany danych

Time	Value in OPC Server	Client Update in "Polling"	Client Update in "Subscription"
1	20	20	20
2	20	20	
3	20	20	
4	20	20	
5	21	21	21
6	21	21	
7	22	22	22



Edit OpcUa-Client node

Delete Cancel Done

Properties

Endpoint: opc.tcp://10.12.0.14:4840

Action: SUBSCRIBE

Interval: 1000 millisecond(s)

Certificate: None, use generated self-signed certificate

Local certificate file, fixed path: Path is not used anymore! Example: selfSigned.p...

Local private key file, fixed path: Path is not used anymore! Example: private_key.f...

Use transport settings: ☐

Max ChunkCount: 1

Max MessageSize: 8192

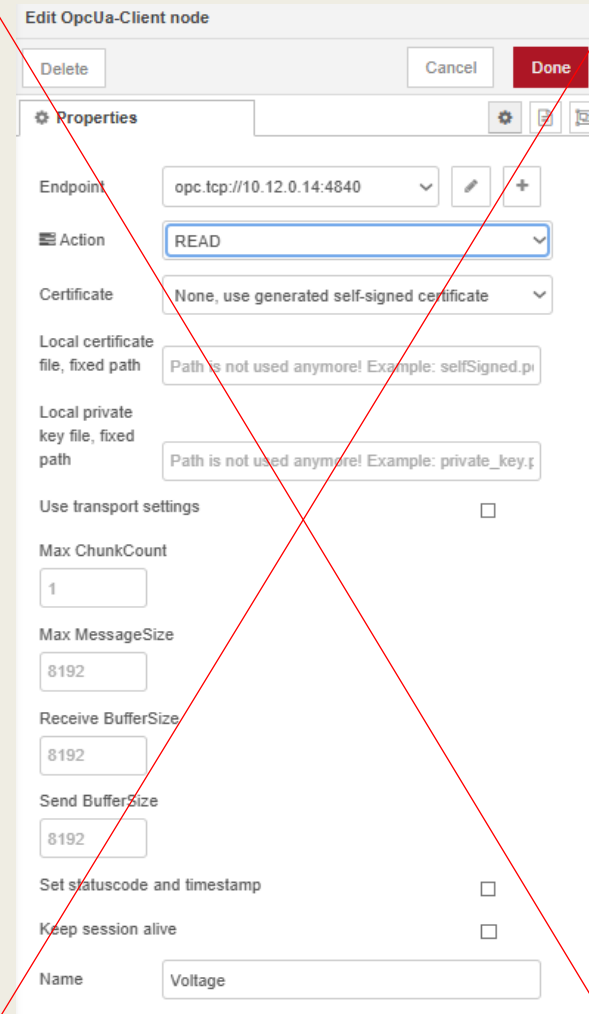
Receive BufferSize: 8192

Send BufferSize: 8192

Set statuscode and timestamp: ☐

Keep session alive: ☐

Name: Voltage



Edit OpcUa-Client node

Delete Cancel Done

Properties

Endpoint: opc.tcp://10.12.0.14:4840

Action: READ

Certificate: None, use generated self-signed certificate

Local certificate file, fixed path: Path is not used anymore! Example: selfSigned.p...

Local private key file, fixed path: Path is not used anymore! Example: private_key.f...

Use transport settings: ☐

Max ChunkCount: 1

Max MessageSize: 8192

Receive BufferSize: 8192

Send BufferSize: 8192

Set statuscode and timestamp: ☐

Keep session alive: ☐

Name: Voltage

Node-Red Dodatkowe funkcje

Edit inject node

Delete Cancel Done

⚙ Properties

👤 Name Name

msg.payload = 0₉

msg.topic = ns=4;s=|var|Delta/x86_64/RTE/SM/CNC/ TV+VV.Application.GVL.Voltage

Edit function node

Delete Cancel Done

⚙ Properties

👤 Name function 4

⚙ Setup On Start On Message On Stop

```
1 msg.payload = Number(msg.payload.toFixed(2));  
2 return msg;
```

Node-Red GitHub

Project Settings

Close

Project

Dependencies

Settings

Files

edit

Packagepackage.json

Flowflows.json

Credentialsflows_cred.json

Encryption disabled

Version Control

Branches

Git remotes

add remote

Project Settings

Close

Project

Dependencies

Settings

IoT-NodeRed-EnergyMeter

Edit project description

Node-RED project for Azure App Service

1.0.0

IoT-NodeRed-EnergyMeter

Edit README.md

PLC

DIADesigner-AX – Pobierz z DIASudio

Aby pobrać DIADesigner-AX, należy zarejestrować się na platformie DIASudio.

DPM

DPMSoft – Pobierz z Delta Download Center

DPMSoft to oprogramowanie do konfiguracji i monitorowania mierników energii Delta DPM.

Node-RED

Node-RED – Pobierz Node-RED

Node-RED to narzędzie do programowania wizualnego, umożliwiające tworzenie aplikacji opartych na zdarzeniach, integrujące urządzenia, API i usługi online.

Aktywacja projektów w Node-RED:

<https://nodered.org/docs/user-guide/projects>

OPC Watch

OPC Watch – Pobierz OPC Watch

OPC Watch to prosty klient OPC UA umożliwiający dostęp do serwerów OPC UA.

VPN

OpenVPN Connect – Pobierz OpenVPN Connect

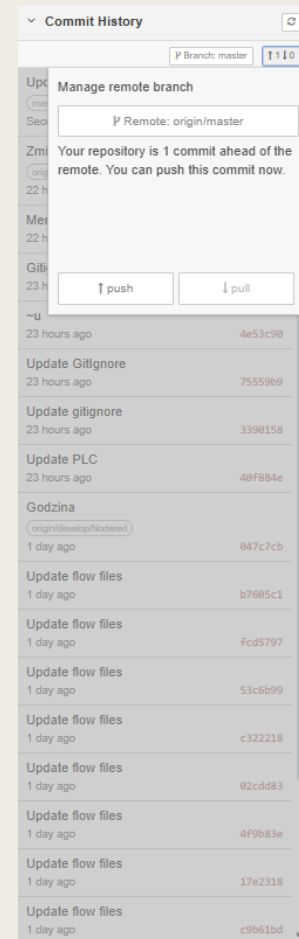
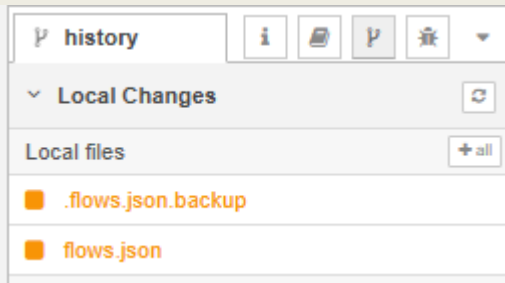
OpenVPN Connect to oficjalny klient OpenVPN, umożliwiający bezpieczne połączenia VPN.

Git

Git for Windows – Pobierz Git

Node-RED – Zależności

Node-Red GitHub - Commity



Node-Red GitHub

Update flow files	RP	rafcel2222-prog	1 day ago	b943ab1
Update flow files	RP	rafcel2222-prog	1 day ago	f841fb2
Update flow files	RP	rafcel2222-prog	1 day ago	3059c78
Update flow files	RP	rafcel2222-prog	1 day ago	4f32820
Update flow files	RP	rafcel2222-prog	1 day ago	9b686cd
Update flow files	RP	rafcel2222-prog	1 day ago	a76e6e3
Update flow files	RP	rafcel2222-prog	1 day ago	489955f
Update server.js Settings.js	PC	PatrykChoinski	6 days ago	1579176
Update admin.auth	PC	PatrykChoinski	6 days ago	5ede400
Poprawki server.js settings.js	PC	PatrykChoinski	6 days ago	e080b3a
Update settings.js	PC	PatrykChoinski	7 days ago	fd13a4f
Update Deployment Gitignore	PC	PatrykChoinski	7 days ago	8b3f779
Update PLC	PC	PatrykChoinski	7 days ago	fdbff94
Update README.md	PC	Patryk Choński	7 days ago	f02d204
Update README.md	PC	Patryk Choński	7 days ago	dc68261
Dodalism6 current	RP	rafcel2222-prog	7 days ago	0dd0a30
Update flow files	RP	rafcel2222-prog	7 days ago	21d541a
Update flow files	RP	rafcel2222-prog	7 days ago	1ac4373
Update flow files	RP	rafcel2222-prog	7 days ago	eaef8ffa
Update flow files	RP	rafcel2222-prog	7 days ago	d958a9f
Update README.md	PC	Patryk Choński	7 days ago	538d626
Update ReadMe	PC	PatrykChoinski	8 days ago	cadb164
yml	PC	PatrykChoinski	8 days ago	a822434
yml	PC	PatrykChoinski	8 days ago	07f3699
Update Deployment	PC	PatrykChoinski	8 days ago	eb4aba8
Stabilny yml	PC	PatrykChoinski	8 days ago	cd96166
.	PC	PatrykChoinski	8 days ago	245903c

Publiczne Repozytorium

- <https://github.com/PatrykChoinski/IoT-NodeRed-EnergyMeter>

The screenshot shows the GitHub repository page for **IoT-NodeRed-EnergyMeter** by user **PatrykChoinski**. The repository is public and has 2 issues, 0 pull requests, 0 projects, 0 wiki pages, 0 security advisories, 0 insights, and 0 settings. The repository has 0 stars, 0 forks, and 0 watchers. The repository is currently on the **master** branch, with 2 branches and 0 tags. The repository contains 131 commits. The repository is currently on the **master** branch, with 2 branches and 0 tags. The repository contains 131 commits. The repository is currently on the **master** branch, with 2 branches and 0 tags. The repository contains 131 commits.

File	Commit Message	Commit Hash	Commit Date
.github/workflows	ym	b50837f	yesterday
Documentation	Docs, Readme PLC		3 weeks ago
PLC	Merge branch 'develop/Nodered'		yesterday
.flows.json.backup	Zmiana rozmieszczen elementow i poprawy nazwenictwa		yesterday
.gitignore	Update Gitignore		yesterday
README.md	Update README.md		last week
flows.json	Zmiana rozmieszczen elementow i poprawy nazwenictwa		yesterday
flows_cred.json	Powrot bez folderu ./nodered		last week
package-lock.json	ij		3 weeks ago
package.json	...		last week
server.js	Update server.js		last week
settings.js	Update server.js		last week

About
No description, website, or topics provided.

Releases
No releases published
[Create a new release](#)

Packages
No packages published
[Publish your first package](#)

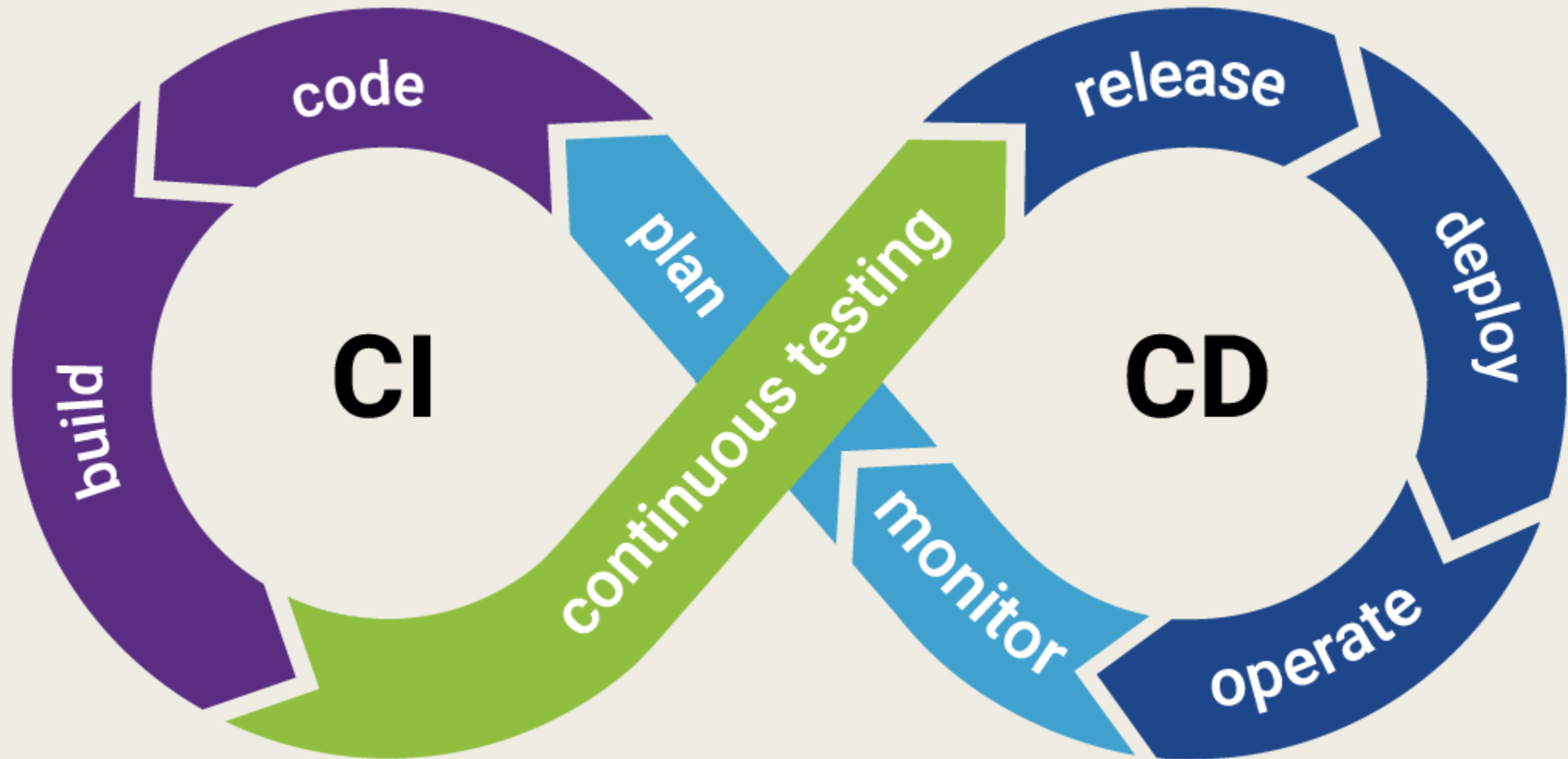
Contributors 2

- rafcel2222-prog**
- PatrykChoinski** Patryk Choiński

KONIEC?

Nie tak prędko!

GitHub Actions i Workflows



GitHub Actions i Workflows

master_iot-nodered-energymeter.yml

```
1  name: Build and deploy Node.js app to Azure Web App - IoT-NodeRed-EnergyMeter
2
3  on:
4    push:
5      branches:
6        - master
7    workflow_dispatch:
8
9  permissions:
10   id-token: write
11   contents: read
12
13  jobs:
14    build:
15      runs-on: ubuntu-latest
16
17      steps:
18        - uses: actions/checkout@v4
19
20        - name: Set up Node.js version
21          uses: actions/setup-node@v3
22          with:
23            node-version: '20.x'
24
25        # ☒ Zmieniono z "npm ci" na "npm install"
26        - name: Install dependencies
27          run: npm install
28
29        - name: Build app
30          run: npm run build --if-present
31
32        - name: Upload artifact for deployment job
33          uses: actions/upload-artifact@v4
34          with:
35            name: node-app
36            path: .
37
38    deploy:
39      runs-on: ubuntu-latest
40      needs: build
41
42      steps:
43        - name: Download artifact from build job
44          uses: actions/download-artifact@v4
45          with:
46            name: node-app
47
48        - name: Login to Azure
```


GitHub Actions i Workflows

← Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter

Update Debug Voltage #64 Cancel workflow ...

Summary

Jobs

- ✓ build
- deploy

Run details

Usage

Workflow file

build
succeeded now in 27s

Search logs

- > ✓ Set up job 1s
- > ✓ Checkout repository 2s
- > ✓ Set up Node.js 1s
- > ✓ Install dependencies 10s
- > ✓ Prepare artifact 8s
- > ✓ Upload artifact 3s
- > ✓ Post Set up Node.js 0s
- ▼ ✓ **Post Checkout repository** 0s
 - 2 /usr/bin/git version
 - 3 git version 2.51.2
 - 4 Temporarily overriding HOME='/home/runner/work/_temp/1bb30e51-9810-4b85-8ebb-d8e3b5alf47f' before making global git config changes
 - 5 Adding repository directory to the temporary git global config as a safe directory
 - 6 /usr/bin/git config --global --add safe.directory /home/runner/work/IoT-NodeRed-EnergyMeter/IoT-NodeRed-EnergyMeter
 - 7 /usr/bin/git config --local --name-only --get-regexp core.sshCommand
 - 8 /usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'core.sshCommand' && git config --local --unset-all 'core.sshCommand' || :"
 - 9 /usr/bin/git config --local --name-only --get-regexp http.https://github.com/.extraheader
 - 10 http.https://github.com/.extraheader
 - 11 /usr/bin/git config --local --unset-all http.https://github.com/.extraheader
 - 12 /usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'http.https://github.com/.extraheader' && git config --local --unset-all 'http.https://github.com/.extraheader' || :"
- > ✓ Complete job 0s

GitHub Actions i Workflows

Actions

New workflow

All workflows

Build and deploy Node-RED app to Azu...

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter #33: Commit [21fe993](#) pushed by [PatrykChoinski](#) 3m 3s

✓ Powrot bez folderu ./nodered

Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter #33: Commit [21fe993](#) pushed by [PatrykChoinski](#) master Oct 26, 6:34 PM GMT+1 3m 50s ...

✓ Update settings.js

Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter #32: Commit [22cb700](#) pushed by [PatrykChoinski](#) master Oct 26, 6:13 PM GMT+1 3m 27s ...

✓ Zmiany w package

Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter #31: Commit [c364e46](#) pushed by [PatrykChoinski](#) master Oct 26, 12:41 PM GMT+1 3m 19s ...

✓ Przygotowanie do działania z folderu ./nodered

Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter #30: Commit [ffc9735](#) pushed by [PatrykChoinski](#) master Oct 26, 12:23 PM GMT+1 3m 42s ...

✗ Enable projects in Node-RED editor theme

Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter #29: Commit [8065667](#) pushed by [PatrykChoinski](#) master Oct 26, 11:19 AM GMT+1 3m 24s ...

✓ Update README with Node-RED projects activation link

Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter #28: Commit [a551888](#) pushed by [PatrykChoinski](#) master Oct 26, 11:18 AM GMT+1 4m 9s ...

✓ yml

Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter #27: Commit [0ff8c8](#) pushed by [PatrykChoinski](#) master Oct 18, 7:22 PM GMT+2 3m 54s ...

✗ yml

Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter #26: Commit [a3041cc](#) pushed by [PatrykChoinski](#) master Oct 18, 7:06 PM GMT+2 3m 40s ...

✗ yml

Build and deploy Node-RED app to Azure Web App - IoT-NodeRed-EnergyMeter #25: Commit [30d0596](#) pushed by [PatrykChoinski](#) master Oct 18, 6:59 PM GMT+2 1m 40s ...

GitHub Actions i Workflows

The screenshot displays a GitHub Actions workflow run interface. On the left, a sidebar shows the workflow steps: 'Summary', 'Jobs', 'Run details', 'Usage', and 'Workflow file'. Under 'Jobs', the 'build' job is selected and marked as failed. The main panel shows the 'build' job details, including a search bar, an 'Explain error' button, and a refresh icon. The job status is 'failed 3 weeks ago in 11s'. The 'Clean install dependencies' step is expanded, showing a list of npm error messages. The errors indicate that the 'npm ci' command cannot be used because there is no existing 'package-lock.json' file. The final error message states: 'Error: Process completed with exit code 1.' Below the error log, a summary of the workflow steps is shown, including 'Build app', 'Upload artifact for deployment job', 'Post Set up Node.js version', 'Post Run actions/checkout@v4', and 'Complete job'.

build
failed 3 weeks ago in 11s

Search logs Explain error

Jobs

- build
- deploy

Run details

- Usage
- Workflow file

Clean install dependencies 3s

```
1 ▶ Run npm ci
4 npm error code EUSAGE
5 npm error
6 npm error The `npm ci` command can only install with an existing package-lock.json or
7 npm error npm-shrinkwrap.json with lockfileVersion >= 1. Run an install with npm@5 or
8 npm error later to generate a package-lock.json file, then try again.
9 npm error
10 npm error Clean install a project
11 npm error
12 npm error Usage:
13 npm error npm ci
14 npm error
15 npm error Options:
16 npm error [--install-strategy <hoisted|nested|shallow|linked>] [--legacy-bundling]
17 npm error [--global-style] [--omit <dev|optional|peer>] [--omit <dev|optional|peer> ...]]
18 npm error [--include <prod|dev|optional|peer>] [--include <prod|dev|optional|peer> ...]]
19 npm error [--strict-peer-deps] [--foreground-scripts] [--ignore-scripts] [--no-audit]
20 npm error [--no-bin-links] [--no-fund] [--dry-run]
21 npm error [-w|--workspace <workspace-name>] [-w|--workspace <workspace-name> ...]]
22 npm error [-ws|--workspaces] [--include-workspace-root] [--install-links]
23 npm error
24 npm error aliases: clean-install, ic, install-clean, isntall-clean
25 npm error
26 npm error Run "npm help ci" for more info
27 npm error A complete log of this run can be found in: /home/runner/.npm/_logs/2025-10-17T13_47_26_193Z-debug-0.log
28 Error: Process completed with exit code 1.
```

Build app 0s

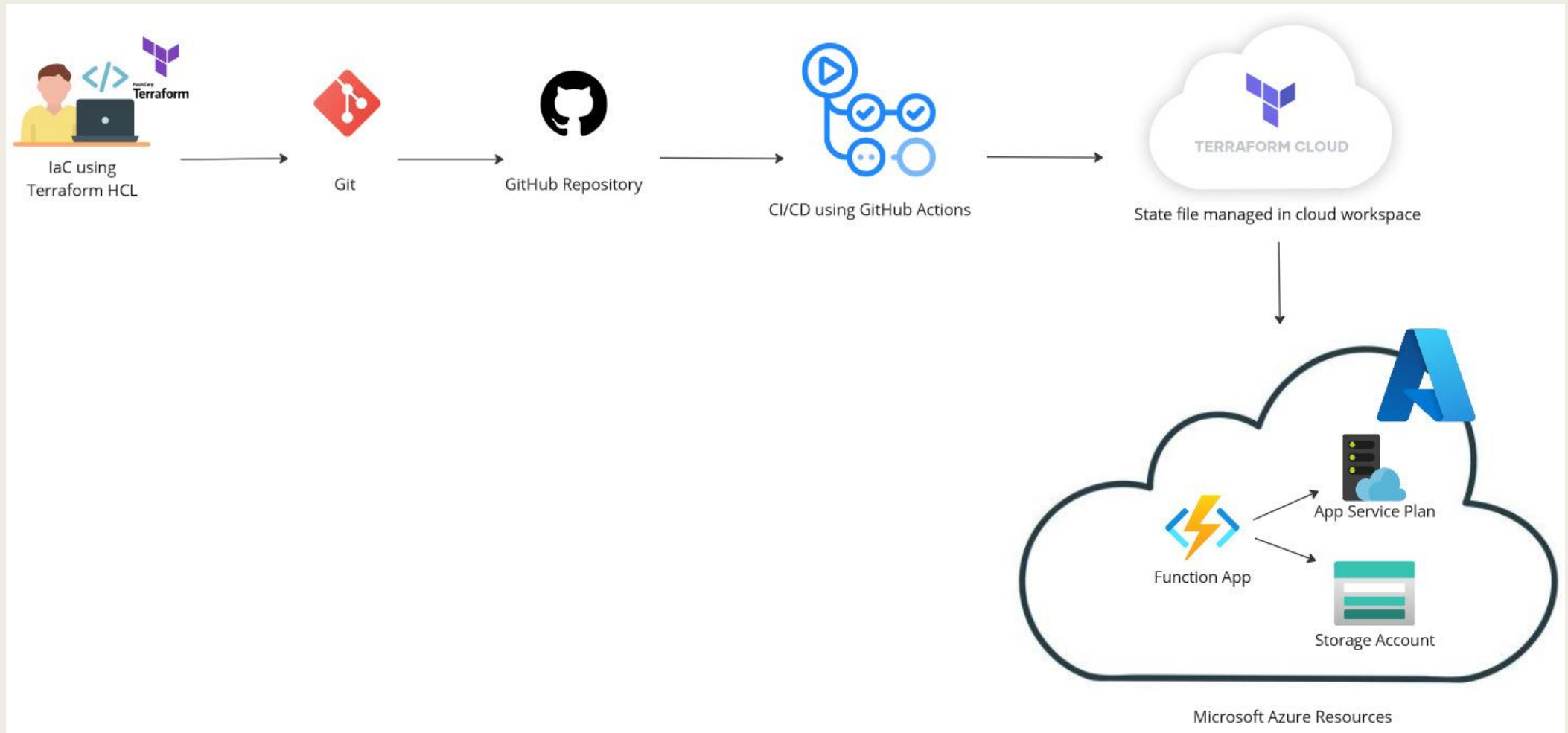
Upload artifact for deployment job 0s

Post Set up Node.js version 0s

Post Run actions/checkout@v4 1s

Complete job 0s

Hosting na Azure



Azure

<https://iot-nodered-energymeter-a0bgd5fhd3cwhsgj.polandcentral-01.azurewebsites.net/red>

<https://iot-nodered-energymeter-a0bgd5fhd3cwhsgj.polandcentral-01.azurewebsites.net/ui>

Create Web App ...

Basics Database Deployment Networking Monitor + secure Tags Review + create

App Service Web Apps lets you quickly build, deploy, and scale enterprise-grade web, mobile, and API apps running on any platform. Meet rigorous performance, scalability, security and compliance requirements while using a fully managed platform to perform infrastructure maintenance. [Learn more](#)

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure for Students

Resource Group *

(New) Resource group

[Create new](#)

Instance Details

Name

Web App name

.azurewebsites.net

☒ Secure unique default hostname on. [More about this update](#)

Publish *

☒ Code ☐ Container

Runtime stack

Loading...

Operating System

☒ Linux ☐ Windows

Region

Loading...

Pricing plans

App Service plan pricing tier determines the location, features, cost and compute resources associated with your app. [Learn more](#)

Linux Plan

Loading...

Review + create

< Previous

Next : Database >

MS Azure

<https://iot-nodered-energymeter-a0bgd5fhd3cwhsgj.polandcentral-01.azurewebsites.net/red>

<https://iot-nodered-energymeter-a0bgd5fhd3cwhsgj.polandcentral-01.azurewebsites.net/ui>

The screenshot displays the Azure portal interface for a web application named "IoT-NodeRed-EnergyMeter". The left sidebar contains navigation links for Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Events (preview), Resource visualizer, Deployment, Settings, Performance, App Service plan, Development Tools, API, Monitoring, Automation, and Support + troubleshooting. The main content area is divided into several sections:

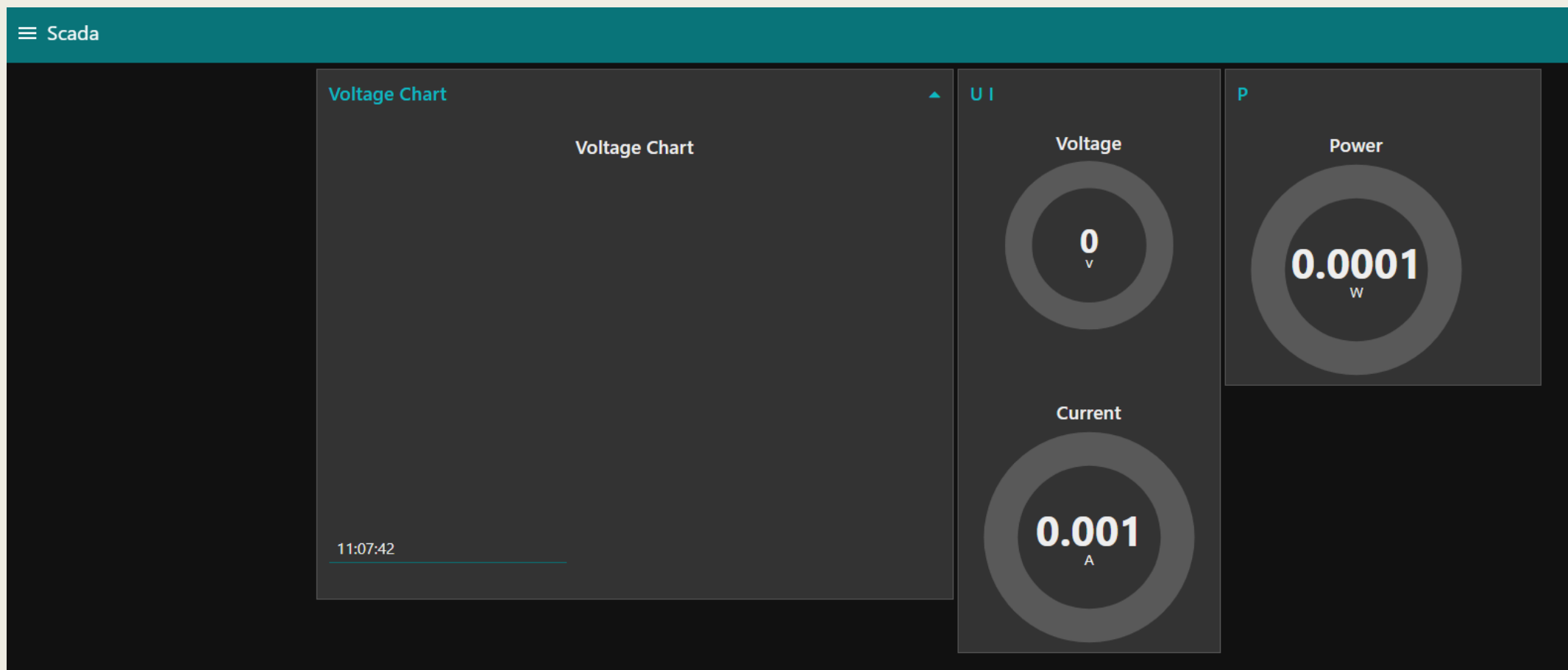
- Essentials:** A summary of key information including Resource group (NodeRed-IoT), Status (Running), Location (Poland Central), Subscription (Azure for Students), and Subscription ID (2eb1f329-4410-4b67-8622-aa9e480fb5f2).
- Properties:** A tabbed view showing details about the Web app, Domains, and Hosting. The Web app section includes Name (IoT-NodeRed-EnergyMeter), Publishing model (Code), and Runtime Stack (Node - 20-lts). The Domains section shows the Default domain (iot-nodered-energymeter-a0bgd5fhd3cwhsgj.polandcentral-01.azurewebsites.net) and a Custom domain (Add custom domain). The Hosting section shows Plan Type (App Service plan), Name (ASP-NodeRedIoT-bcde), Operating System (Linux), and Instance Count (1).
- Deployment Center:** A section for managing deployments, showing the last deployment was successful on Monday, 3 October, at 12:45:13 PM.
- Application Insights:** A section for monitoring application performance, showing the Name (IoT-NodeRed-EnergyMeter) and Region (polandcentral).
- Networking:** A section for managing network resources, showing the Virtual IP address (20.215.12.2) and Outbound IP addresses (20.215.90.80, 20.215.90.87, 20.215.88.5...).

At the bottom of the page, there is a note: "Add or remove favorites by pressing Ctrl+Shift+F".

Dostęp do logów

```
2025-11-04T10:05:35.3351319Z Completed successfully in 7 seconds
2025-11-04T10:05:52.2738417Z
2025-11-04T10:05:52.2833114Z
2025-11-04T10:05:52.2833322Z
2025-11-04T10:05:52.2833389Z
2025-11-04T10:05:52.2833411Z
2025-11-04T10:05:52.2833432Z
2025-11-04T10:05:52.2833466Z A P P   S E R V I C E   O N   L I N U X
2025-11-04T10:05:52.2833485Z
2025-11-04T10:05:52.2833514Z Documentation: http://aka.ms/webapp-linux
2025-11-04T10:05:52.2833534Z NodeJS quickstart: https://aka.ms/node-qs
2025-11-04T10:05:52.2833574Z NodeJS Version : v20.19.3
2025-11-04T10:05:52.2833595Z Note: Any data outside '/home' is not persisted
2025-11-04T10:05:52.2833614Z
2025-11-04T10:05:54.9428854Z Starting OpenBSD Secure Shell server: sshd.
2025-11-04T10:05:55.020268Z WEBSITES_INCLUDE_CLOUD_CERTS is not set to true.
2025-11-04T10:05:55.0709902Z Updating certificates in /etc/ssl/certs...
2025-11-04T10:06:09.5611987Z rehash: warning: skipping ca-certificates.crt,it does not contain exactly one certificate or CRL
2025-11-04T10:06:09.6794544Z 4 added, 0 removed; done.
2025-11-04T10:06:09.6888406Z Running hooks in /etc/ca-certificates/update.d...
2025-11-04T10:06:09.7016391Z done.
2025-11-04T10:06:09.7318575Z CA certificates copied and updated successfully.
2025-11-04T10:06:10.0021882Z Starting periodic command scheduler: cron.
2025-11-04T10:06:10.1838799Z Found build manifest file at '/home/site/wwwroot/oryx-manifest.toml'. Deserializing it...
2025-11-04T10:06:10.2020145Z Could not find operation ID in manifest. Generating an operation id...
2025-11-04T10:06:10.2020525Z Build Operation ID: 11656d23-10a4-4800-a461-cb6a0519031e
2025-11-04T10:06:10.504627Z Environment Variables for Application Insight's IPA Codeless Configuration exists..
2025-11-04T10:06:10.5544628Z Writing output script to '/opt/startup/startup.sh'
2025-11-04T10:06:10.6358449Z Running #!/bin/sh
2025-11-04T10:06:10.6358989Z
2025-11-04T10:06:10.6359101Z # Enter the source directory to make sure the script runs where the user expects
2025-11-04T10:06:10.6359135Z cd "/home/site/wwwroot"
2025-11-04T10:06:10.6359154Z
2025-11-04T10:06:10.6359176Z export NODE_PATH=/usr/local/lib/node_modules:$NODE_PATH
2025-11-04T10:06:10.6359196Z if [ -z "$PORT" ]; then
2025-11-04T10:06:10.6359216Z     export PORT=8080
2025-11-04T10:06:10.6359236Z fi
```

Jak połączyć Cloud Azure z Siecią PLC/OPC?



Azure Virtual Network Gateway

[Home](#) > [Create a resource](#) > [Marketplace](#) >

Create virtual network gateway

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources. ⓘ

Subscription * ▼ Azure for Students

Resource group ⓘ Select a virtual network to get resource group

Instance details

Name *

Region * ▼ Poland Central
[Deploy to an Azure Extended Zone](#) ⓘ

Gateway type * ⓘ ☒ VPN ☐ ExpressRoute

SKU * ⓘ ▼ VpnGw2AZ

Generation ⓘ ▼ Generation2

Enable Advanced Connectivity ⓘ ☐ Enabled ☒ Disabled

Virtual network * ⓘ ▼ [Create virtual network](#)

ⓘ Only virtual networks in the currently selected subscription and region are listed.

Public IP address

Public IP address * ⓘ ☒ Create new ☐ Use existing

Public IP address name *

Public IP address SKU Standard

Assignment ☐ Dynamic ☒ Static

[Review + create](#) [Previous](#) [Next : Tags >](#) [Download a template for automation](#)

NodeRedVPNGateway

Virtual network gateway

[Diagnose my virtual network gateway](#) [Analyze traffic through network gateway](#) [Analyze performance for this virtual network gateway](#)

[Refresh](#) [Move](#) [Delete](#)

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer

Essentials

Resource group (move)	: NodeRed-IoT	SKU	: VpnGw1AZ
Location	: Poland Central	Gateway type	: VPN
Subscription (move)	: Azure for Students	VPN type	: Route-based
Subscription ID	: 2eb1f329-4410-4b67-8622-aa9e480fb5f2	Virtual network	: VNet
Tags (edit)	: Add tags	Public IP address	: 134.112.129.13 (PLC-VPN-PublicIP)

Point To Site Configuration

 **NodeRedVPNGateway** | Point-to-site configuration ☆ ...

Virtual network gateway

⌵ ⏪

Save

Discard

Delete

Download VPN client

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Configuration

Connections

Point-to-site configuration

Maintenance

Properties

Locks

Monitoring

Automation

Help

Address pool *

10.10.0.0/24 ✓

Tunnel type

OpenVPN (SSL) ▾

Authentication type

Azure certificate ▾

Root certificates

Name	Public certificate data
AzureRootCert	MIIC5TCCAc2gAwIBAgIQbZmFI8hw6LNECsaUSLEICjANBgkqhkiG9w0BAQsFADAVMRMwEQ...

Revoked certificates

Name	Thumbprint

Additional routes to advertise

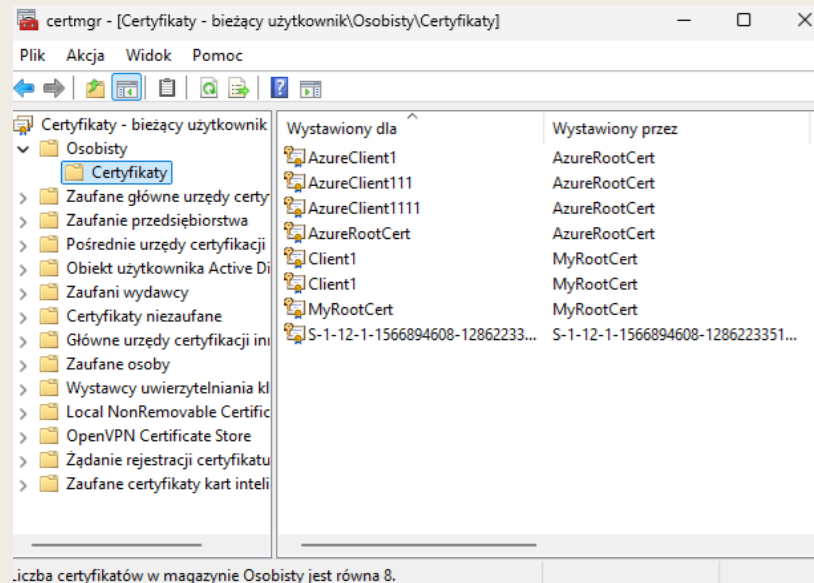
✓

Generowanie Certyfikatu dla Azure

```
PowerShell 7 (x64)
PowerShell 7.5.4
PS C:\Users\InduTech> $rootCert = Get-ChildItem -Path "Cert:\CurrentUser\My" | Where-Object { $_.Subject -like "*AzureRootCert*" }
PS C:\Users\InduTech>
PS C:\Users\InduTech> $clientCert = New-SelfSignedCertificate -Type Custom `
>>     -DnsName "AzureClient1" `
>>     -KeySpec Signature `
>>     -Subject "CN=AzureClient1" `
>>     -KeyExportPolicy Exportable `
>>     -KeyLength 2048 `
>>     -CertStoreLocation "Cert:\CurrentUser\My" `
>>     -Signer $rootCert
```

Generowanie Certyfikatu dla Azure

```
PowerShell 7 (x64)
PowerShell 7.5.4
PS C:\Users\InduTech> $rootCert = Get-ChildItem -Path "Cert:\CurrentUser\My" | Where-Object { $_.Subject -like "*AzureRootCert*" }
PS C:\Users\InduTech>
PS C:\Users\InduTech> $clientCert = New-SelfSignedCertificate -Type Custom `
>> -DnsName "AzureClient1" `
>> -KeySpec Signature `
>> -Subject "CN=AzureClient1" `
>> -KeyExportPolicy Exportable `
>> -KeyLength 2048 `
>> -CertStoreLocation "Cert:\CurrentUser\My" `
>> -Signer $rootCert
```



Eksport klucza do postaci nie binarnej

← Kreator eksportu certyfikatów

Format pliku eksportu
Certyfikaty mogą być eksportowane w wielu różnych formatach plików.

Wybierz format, którego chcesz użyć:

- ☐ Certyfikat X.509 szyfrowany binarnie algorytmem DER (CER)
- ☒ Certyfikat X.509 szyfrowany algorytmem Base-64 (CER)
- ☐ Standard składni wiadomości kryptograficznych — certyfikaty PKCS #7 (P7B)
 - ☐ Jeśli jest to możliwe, dołącz wszystkie certyfikaty ze ścieżki certyfikacji
- ☐ Wymiana informacji osobistych — PKCS #12 (PFX)
 - ☐ Jeśli jest to możliwe, dołącz wszystkie certyfikaty do ścieżki certyfikacji
 - ☐ Usuń klucz prywatny, jeśli eksport został zakończony pomyślnie
 - ☐ Eksportuj wszystkie właściwości rozszerzone
 - ☐ Włącz funkcję prywatności certyfikatu
- ☐ Magazyn certyfikatów seryjnych firmy Microsoft (SST)

Dalej Anuluj

← Kreator eksportu certyfikatów

Eksportowanie klucza prywatnego
Możesz wybrać eksport klucza prywatnego razem z certyfikatem.

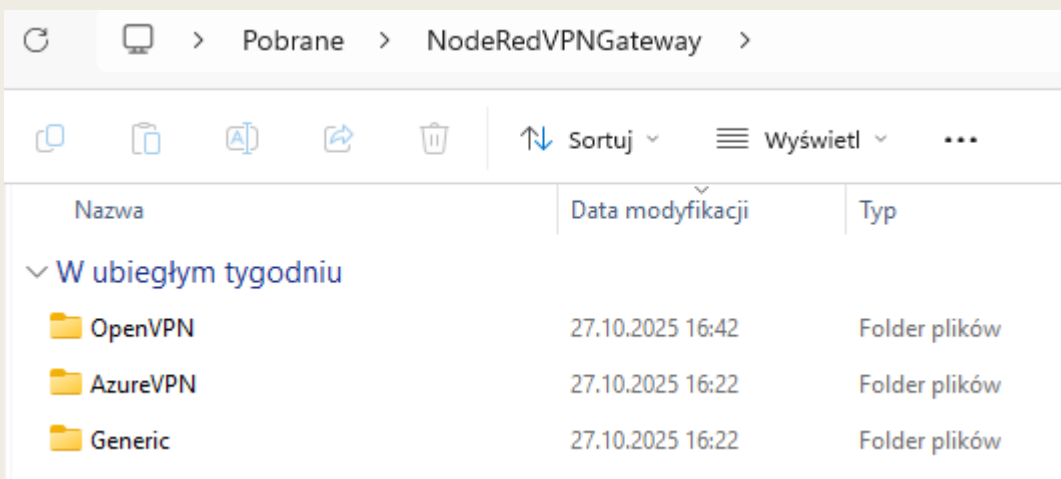
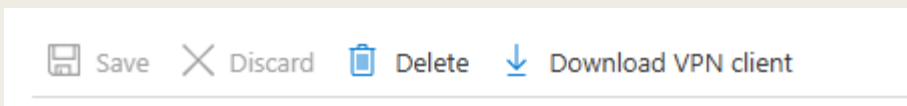
Klucze prywatne są chronione hasłem. Aby wyeksportować klucz prywatny z certyfikatem, musisz wpisać hasło na jednej z kolejnych stron.

Czy chcesz wyeksportować klucz prywatny wraz z certyfikatem?

- ☐ Tak, eksportuj klucz prywatny
- ☒ Nie eksportuj klucza prywatnego

```
1 |-----BEGIN CERTIFICATE-----
2 |MIIC5TCCAc2gAwIBAgIQbZmFI8hw6LNECsaUSLElCjANBgkqhkiG9w0BAQsFADAV
3 |MRMwEQYDVQQDDApNeVJvb3RDZXJ0MB4XDTE1MTAyNzE0NTUxNloXDTE1MTAyNzE1
4 |MTUxNlowFETMBEGA1UEAwKTX15b290Q2VydDCCASIwDQYJKoZIhvcNAQEBBQAD
5 |ggEPADCCAQoCggEBAMu1hJ5Ijzfy4UpE4Gosnt51318ywOj7Gdf1M46S82wQd5jp
6 |GQxZh1Zdf/v3jALyfnI5omCWvssUqbconzXEUt79pY1m1xGuX05x1bNrXmskH29J
7 |i1sasoa7sYQTih6nBnMEAlcNY1D25x/xH/2eOGVJRvnpfjsAuR05AT9Wq9Sg0Ivx
8 |FehzvKI3N/NzK+FsjRNmIf2fBYzkGMwUS2PzWQQC1qKWjwBCHO+91K6T4wRfLmXm
9 |I5ZY/o95qc0iryiUOCDAadx0AsZ4NKZm+FY/0ccF8MvQFaMdQgwhdB/VAYowIHbQ
10 |kuqCspWJLxKlINhJ6kUpODKcJs0kgmAqfc8Wl2ECAwEAAmMCM8wDgYDVVR0PAQH/
11 |BAQDAgKEMB0GA1UdDgQWBBQZ8fBD9zt/v80/Qzj06BjRFOD3xjANBgkqhkiG9w0B
12 |AQsFAAOCAQEAsAXYv6Zki9ofaznEHV/15kjpIbhKaG3lo7gpR0sWhm3Aq6cosPTT
13 |whRlH6bgtbmOBo8gPULN4kEstkyV6h5D3lL+drNdwx0v0zAeztr//Qi58QiV4Cx
14 |id5Ejfd940B0WKYu5u66xjd1TiQZUED8XZQsTB9oLGGYym+wTAHm3JJ5u1DeSD/T
15 |l95OFJLrfxeJcYTosJh9DA1RPQ/YMHGgMNgbcTm319Lezlu9imG2IfICL7fLtc3A
16 |kKs9DKU1zOdDY3Hb+bmzPWlPq+8u72KY4MYoNMOK6rk8wA06+Kga7IBWFMuKcrrD
17 |/Lke+dt7jPSh0h1K49iBj77ZvufZzh4/AA==
18 |-----END CERTIFICATE-----
```

Pobranie konfiguracji ovpn



Dodanie certyfikatu i klucza prywatnego na bazie certyfikatu publicznego

Plik openvpn - .ovpn nie jest kompletny dopóki nie wpiszę mu certyfikatu i klucza prywatnego, który możemy wygenerować na bazie certyfikatu publicznego, który udostępniliśmy do Azure Virtual Gateway Network

  Kreator eksportu certyfikatów

Eksportowanie klucza prywatnego

Możesz wybrać eksport klucza prywatnego razem z certyfikatem.

Klucze prywatne są chronione hasłem. Aby wyeksportować klucz prywatny z certyfikatem, musisz wpisać hasło na jednej z kolejnych stron.

Czy chcesz wyeksportować klucz prywatny wraz z certyfikatem?

☒ Tak, eksportuj klucz prywatny

☐ Nie eksportuj klucza prywatnego

[illegible]

DZIĘKUJEMY ZA UWAGĘ

To już koniec!